



UNIVERSITÀ DI PISA

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

Candidato:

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Relatore:

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Outline

1. Cuprates and Hubbard physics
2. The Hartree-Fock method
3. The normal phase
4. The antiferromagnetic phase
5. The superconducting phase
6. Conclusions and outlook

Cuprates and Hubbard physics

The timeline of superconductivity

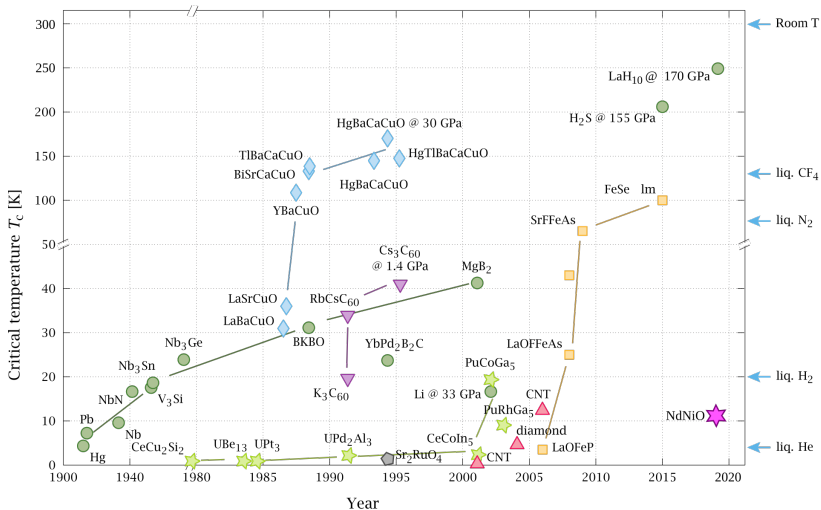


Figure 1: Timeline of the discoveries of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to [the license CC BY-SA 4.0](#).

The timeline of superconductivity

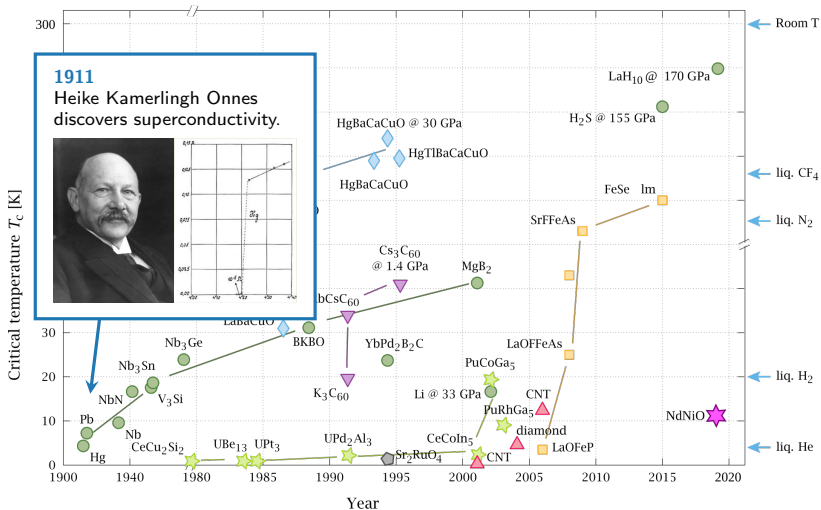


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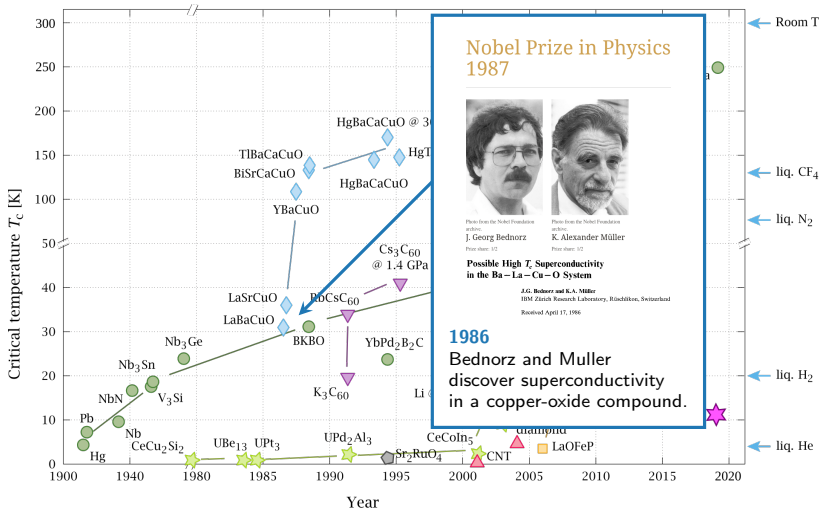


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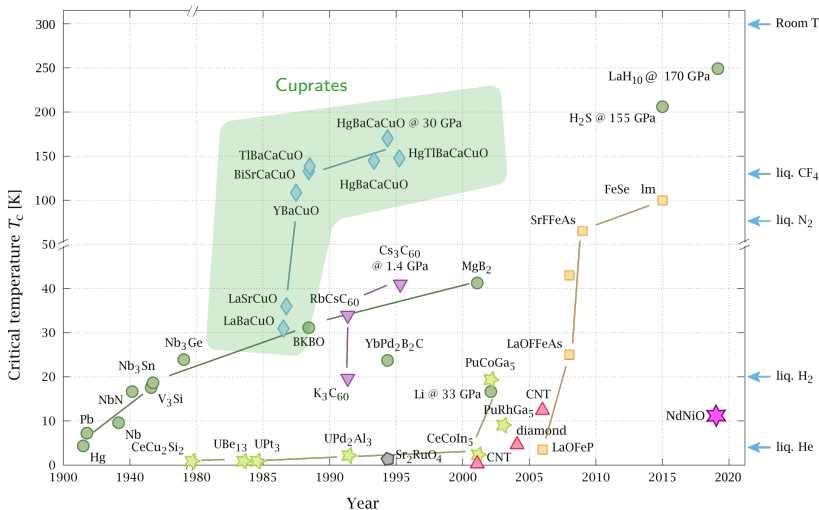


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The phase diagram of cuprates

Cuprates are the prototypical example of **strongly correlated materials**:

- ◇ **Antiferromagnetism** at null and low doping;
- ◇ ***d*-wave superconducting** domes.

Both arise from **strong electronic correlations**.



Failure of band theory in microscopic description.

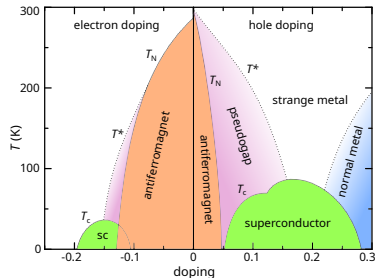


Figure 2: Schematic phase diagram of cuprates.
Author: Holger Motzkau, [Wikimedia commons](#),
distributed according to [the license CC BY-SA 4.0](#)

These materials are stackings of:

- ◇ 2D copper oxide active layers;
- ◇ intermediate layers of inert material.

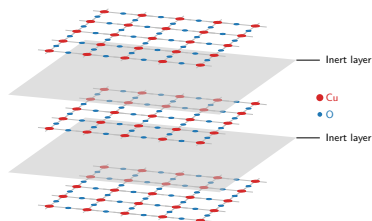


Figure 3: Sketch of a typical cuprate.

The single-band Hubbard model on the square lattice

$$\hat{H}_{\text{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \quad \text{with } t, U > 0.$$

$$\hat{H}_{\text{HM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \quad \text{with } t, U > 0.$$

Despite its simplicity, the Hubbard model describes:

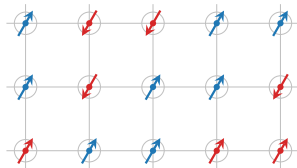
- ◊ Insulation at half-filling (**Mott localisation**);
- ◊ **Antiferromagnetism** induced by Mott localisation;
- ◊ ***d*-wave superconductivity** emerging by doping the antiferromagnetic phase.

What is the origin of the insulating state at half-filling?

The Hubbard model at strong coupling,

$$t \ll U,$$

describes the **Mott insulator** state.

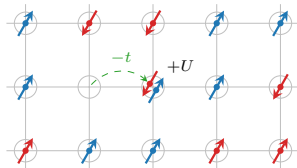


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Mott localisation: a many-body insulator

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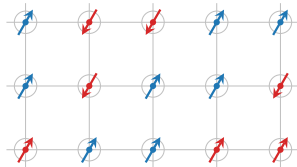
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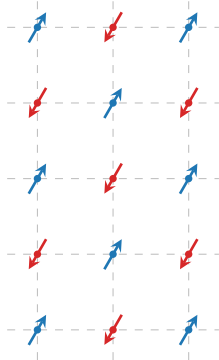


Mott insulation mechanism

Electrons are **localised** as in a sort of “traffic jam”.

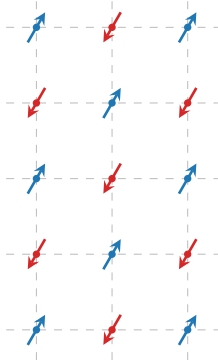


What is the origin of the antiferromagnetic ordering?



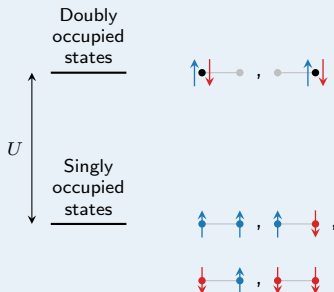
Antiferromagnetism in the undoped Hubbard model

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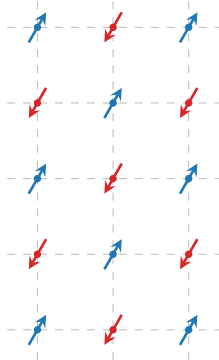


Example: 2 sites, 2 electrons

At $U \gg t$, singlet pairing is energetically favoured:

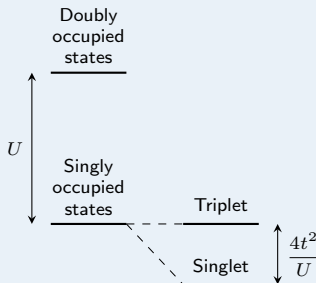


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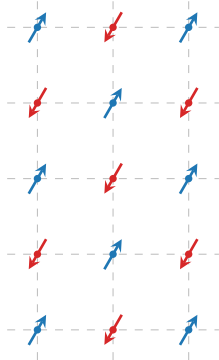
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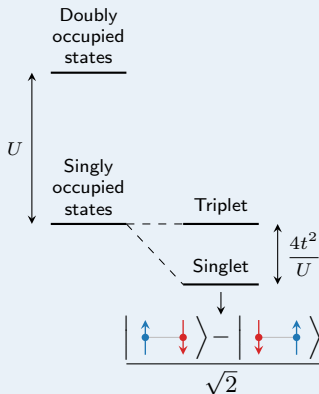
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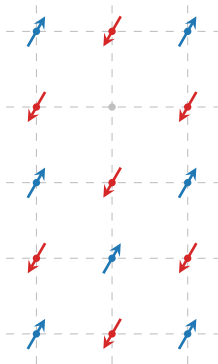


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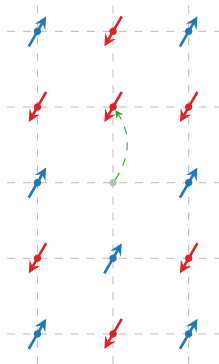
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What attractive interaction originates superconductivity when doping the AF state in the Hubbard model?

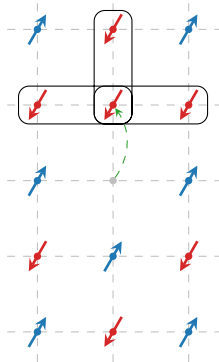


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--- Kinetic processes

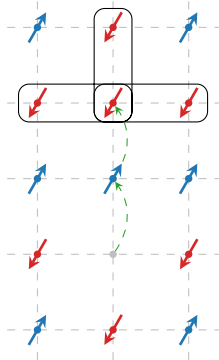
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--- Kinetic processes

Local triplets

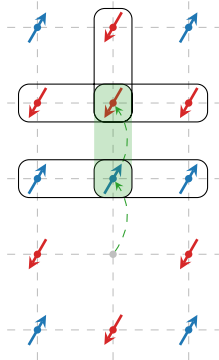
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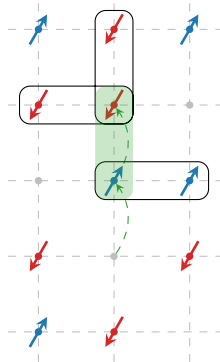


--- Kinetic processes

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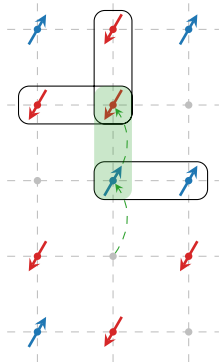
Local AF ordering

Neighbouring electrons with opposite spin
“move together”:



Effective attractive interaction...

What attractive interaction originates superconductivity when doping the AF state in the Hubbard model?



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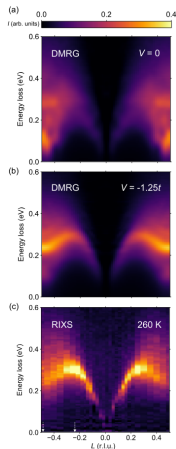
Local AF ordering

Neighbouring electrons with opposite spin
“move together”:



Effective attractive interaction... favoured by doping.

Recent developments: the extended Hubbard model



Observed: incompatibilities of the Hubbard model with the spectrum of magnetic excitation in cuprate $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$.

Figure 5: Image source: Hari Padma et al., 2025. *Physical Review X* 15 (2): 021049.

Recent developments: the extended Hubbard model

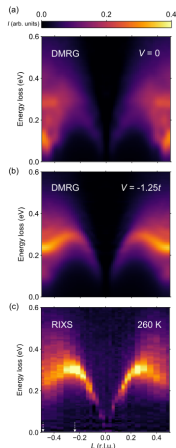


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Observed: incompatibilities of the Hubbard model with the spectrum of magnetic excitation in cuprate $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$.



Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\text{EHM}} = \hat{H}_{\text{HM}} - V \underbrace{\sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V}.$$

$\hat{H}_V$ can act as an independent source of *d*-wave superconductivity.

The model (EHM):

$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} - \underbrace{V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \quad \text{with } t, U, V > 0.$$

The motivation: compatibility with some aspects of cuprates physics, including d -wave superconductivity.

The target: understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

The method: Hartree-Fock variational method, at low temperature.

The Hartree-Fock method

The constrained minimisation framework

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0 .

Variational principle:

$$\forall |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} : E[\Phi] \geq E_0.$$

$\Downarrow$

Constrained minimisation procedure

$$|\Psi\rangle : \left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi\rangle} = 0.$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.

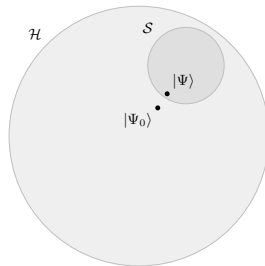


Figure 6: Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

The Hartree-Fock subspace of states

$\mathcal{S}$ is the set of all **gaussian states**,

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle, \quad \left\{ \hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger} \right\} = \delta_{\alpha\beta},$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

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with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.



Each gaussian state can be identified **uniquely** by fixing

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle, \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle.$$

These are the **variational parameters** collected in the vector $\mathbf{v}$. We aim to find the optimal $\mathbf{v}$, such that

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle.$$

Phase transition: spontaneous breaking of some symmetries of the model.

Phase X : shows symmetries $Y, Z \dots$



Restrict to $\mathcal{P} \subset \mathcal{S}$, gaussian states with symmetries $Y, Z \dots$



Symmetry selection rules on $\mathbf{v}$ entries.

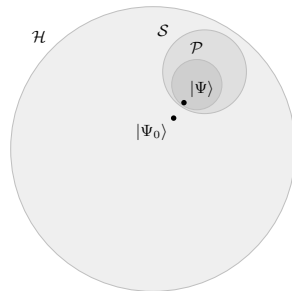
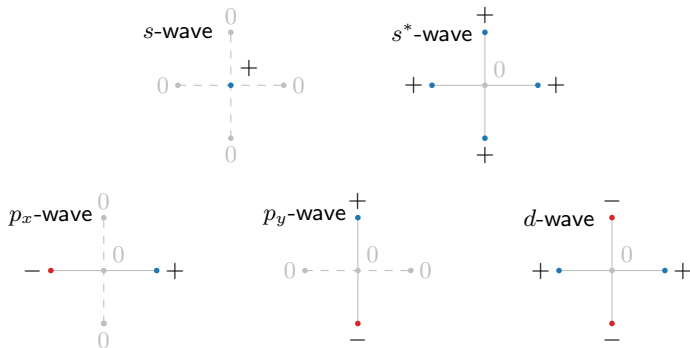


Figure 7: Sketch of the Hilbert space $\mathcal{H}$, the subset $\mathcal{S}$ of gaussian states and the subset $\mathcal{P}$.

Identify symmetries $\rightarrow$ Derive $\mathbf{v}$ $\rightarrow$ Solve the variational problem.

The spatial symmetry channels

It is useful to introduce five spatial symmetry channels:



Function f on NNs: decomposition in harmonics $f^{(\gamma)}$.

The normal phase

The variational problem for the normal phase

Normal phase: no broken symmetries. Just one variational parameter,

$$\mathbf{v} = \left[u^{(s^*)} \right] ,$$

where $u^{(s^*)}$ is the s^* -wave component of $u_{\mathbf{k}\sigma} \equiv \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle$.

$\Downarrow$

Renormalisation of kinetics

Hopping amplitude is **isotropically renormalised**,

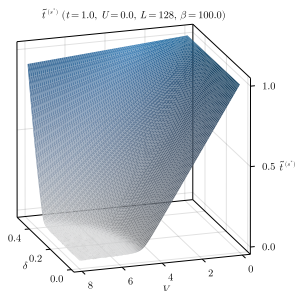
$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} .$$

We have proved the **constraint**:

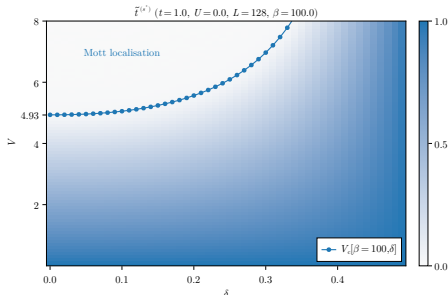
$$0 < \tilde{t}^{(s^*)} < t \quad \implies \quad \text{Possible Mott localisation.}$$

Complete flattening of bare bands

In (δ, V) plane, we have determined theoretically the presence of a Mott critical boundary $V_c[\beta, \delta]$,



(a) Numerical results for $\tilde{t}^{(s^*)}$.



(b) Critical Mott boundary.

Within the Mott plateau, electrons are completely localised and **bands are flattened**.

The role of V/t :

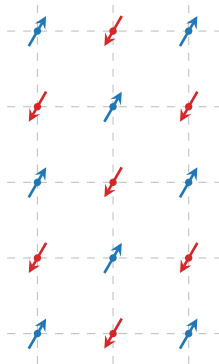
- ◇ **Renormalise hopping;**
- ◇ **Induce complete Mott localisation:**

Effective kinetics energy scale $\ll$ Interaction energy scale .

The role of U/t :

- ◇ At HF level, U does not produce Mott localisation.

The antiferromagnetic phase



In cuprates, the AF phase is **commensurate**:

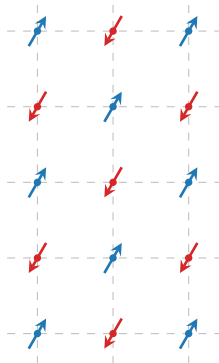
- ◇ 2-sites periodic in each direction;
- ◇ Sublattices oppositely magnetised;
- ◇ Zero net magnetisation.



Staggered magnetisation m :

$$2m = |\langle \hat{n}_{i\uparrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle|,$$

with i a site.



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- ◇ 2-sites periodic in each direction;
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Staggered magnetisation m :

$$2m = |\langle \hat{n}_{i\uparrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle|,$$

with i a site.

Half-filling restriction

We limit to $\delta = 0$ (half-filling) due to **stability considerations**.

Variational parameters for the AF phase:

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix},$$

with $u^{(s^*)}$ defined as for the normal phase.

Renormalisation of kinetics

Mott localisation effect **formally** identical to the normal phase:

$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2}.$$

We have proved the **constraint**: $0 < \tilde{t}^{(s^*)} < t$.

The cooperation on magnetisation

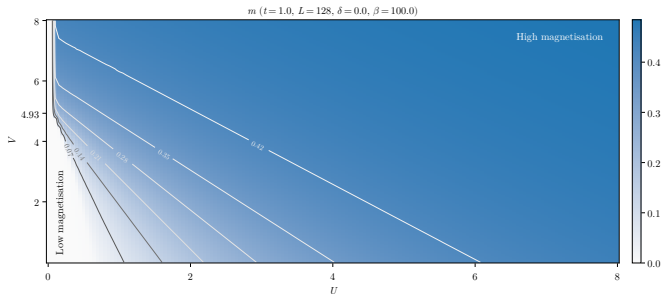


Figure 9: Numerical results for m . The gray curves indicate isovalued lines.

The cooperation on magnetisation

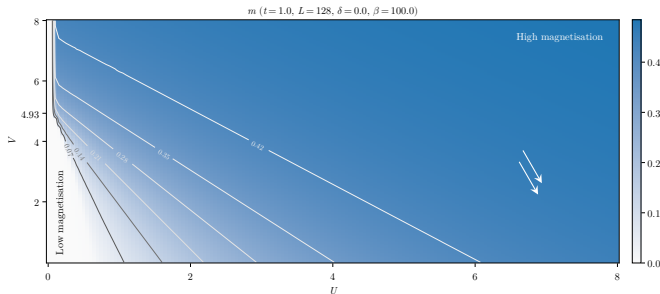


Figure 9: Numerical results for m . The gray curves indicate isovalues lines.

$$U \gg V, t$$

“Conventional” Hubbard magnetisation, induced by the ratio U/t .

The cooperation on magnetisation

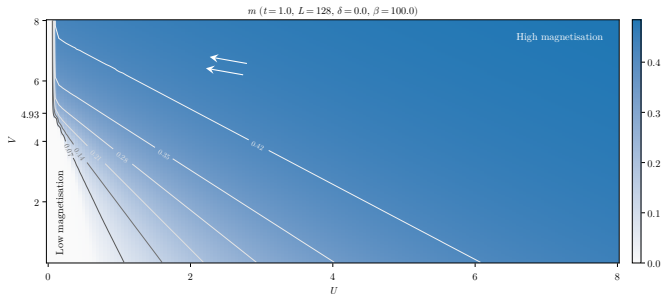
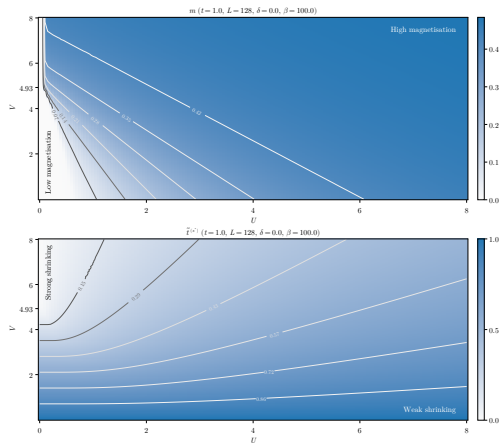


Figure 9: Numerical results for m . The gray curves indicate isovalues lines.

$$V \gg U, t$$

New “unconventional” source of magnetisation, induced by the ratio V/t .

The pairing enhancement of magnetisation



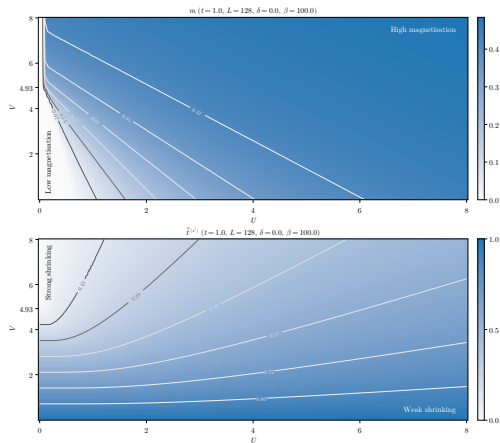
At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}(r^2),$$

where $r = \tilde{t}^{(s^*)}/U$.

Figure 10: Numerical results for m (top) and $\tilde{t}^{(s^*)}$ (bottom). The gray curves indicate isovalues.

The pairing enhancement of magnetisation



At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}(r^2),$$

where $r = \tilde{t}^{(s^*)}/U$.



To enhance m , either:

- ◇ Increase U ;
- ◇ **Decrease $\tilde{t}^{(s^*)}$** .

Figure 10: Numerical results for m (top) and $\tilde{t}^{(s^*)}$ (bottom). The gray curves indicate isovalued lines.

The role of V/t :

- ◇ **Renormalise hopping;**
- ◇ **Magnetise the lattice using Mott physics;**

The role of U/t :

- ◇ Magnetise the lattice, conventional Hubbard physics.



Cooperation among interactions to magnetise the lattice.

The superconducting phase

Cooper pairing:

- ◇ Described by a SC gap $\tilde{\Delta}_{ij}$, roughly the “binding energy” of the pairs;
- ◇ The pair spin obeys sum rules:

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1 = \text{singlet} \oplus \text{triplet}.$$

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The singlet channel

The gap function is inversion symmetric, i.e. its harmonics are

$$\tilde{\Delta}^{(s)}, \quad \tilde{\Delta}^{(s^*)} \quad \text{and} \quad \tilde{\Delta}^{(d)}.$$

We limit to the **singlet channel**. Let the quantities:

$$u_{\mathbf{k}\sigma} \equiv \left\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \right\rangle, \quad w_{\mathbf{k}} \equiv \left\langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow}^\dagger \right\rangle.$$

Variational parameters: symmetric harmonics,

$$\mathbf{v} = \left(u^{(s^*)}, u^{(d)}, w^{(s)}, w^{(s^*)}, w^{(d)} \right)$$

The role of each parameter

- ◇ $u^{(s^*)}, u^{(d)}$: renormalisation of kinetics;
- ◇ $w^{(s)}, w^{(s^*)}$ and $w^{(d)}$: components of the SC gap.

Isotropic gap components:

$$\tilde{\Delta}^{(s^*)} = Vw^{(s^*)} \quad (\text{Mediated by the isotropic non-local attraction})$$

$$\tilde{\Delta}^{(s)} = -Uw^{(s)} \quad (\text{Mediated by the **local repulsion**})$$

Anisotropic gap component:

$$\tilde{\Delta}^{(d)} = Vw^{(d)} \quad (\text{Mediated by the isotropic non-local attraction})$$

The competition among gap components

Isotropic gap components:

$$\tilde{\Delta}^{(s^*)} = Vw^{(s^*)} \quad (\text{Mediated by the isotropic non-local attraction})$$

$$\tilde{\Delta}^{(s)} = -Uw^{(s)} \quad (\text{Mediated by the **local repulsion**})$$

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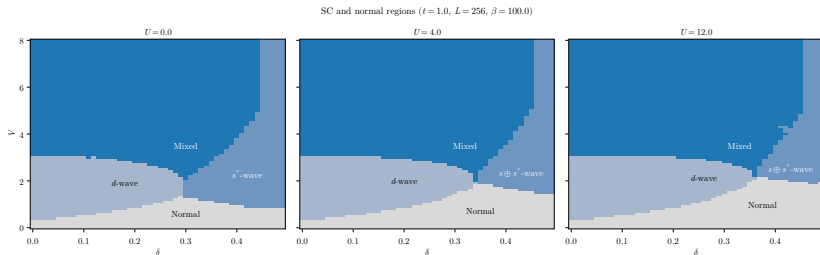


Figure 11: Phase diagram for the SC phase.

In the SC phase, the renormalisation of t gives two components,

$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} \quad \text{and} \quad \tilde{t}^{(d)} \equiv -\frac{u^{(d)}V}{2} .$$

The nematic order parameter

In the SC phase, the renormalisation of t gives two components,

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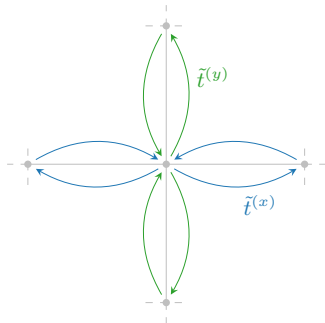


Figure 12: Sketch of nematic hopping.

Anisotropic hopping

Direction-resolved effective hopping:

$$\tilde{t}^{(x)} \equiv \frac{\tilde{t}^{(s^*)} + \tilde{t}^{(d)}}{2},$$

$$\tilde{t}^{(y)} \equiv \frac{\tilde{t}^{(s^*)} - \tilde{t}^{(d)}}{2}.$$

$\Downarrow$

$\tilde{t}^{(d)} \neq 0 \implies$ **nematic SSB.**

Symmetry mixing and insurgence of nematicity

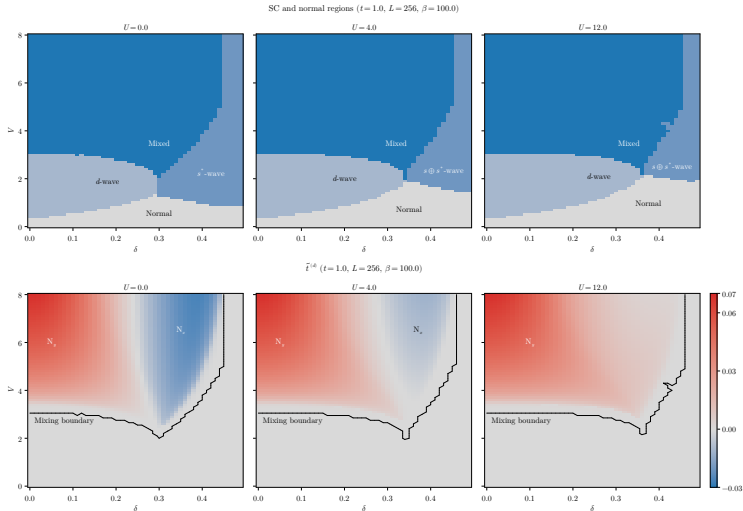


Figure 13: Comparison of phase diagrams for the singlet SC phase and convergence values for the nematic order parameter.

Symmetry mixing and insurgence of nematicity

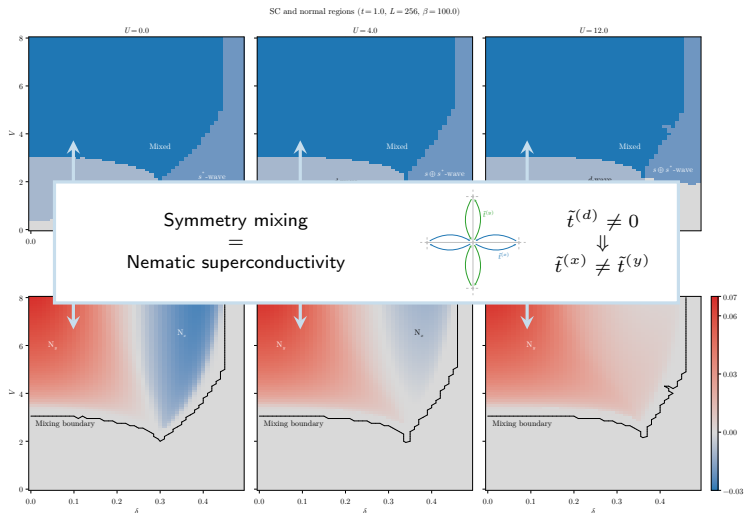


Figure 13: Comparison of phase diagrams for the singlet SC phase and convergence values for the nematic order parameter.

Symmetry mixing and insurgence of nematicity

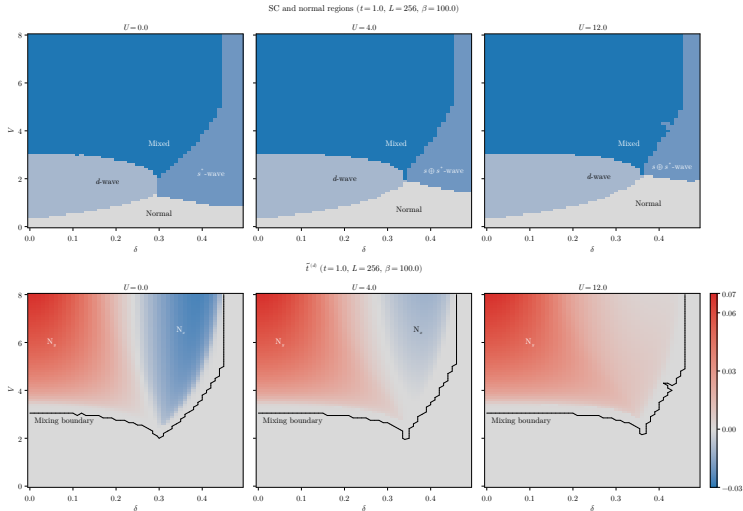


Figure 13: Comparison of phase diagrams for the singlet SC phase and convergence values for the nematic order parameter.

The role of V/t :

- ◇ **Renormalise hopping anisotropically;**
- ◇ **Induce mixed-symmetry superconductivity;**
- ◇ **Induce nematic symmetry breaking.**

The role of U/t :

- ◇ **Participate in superconductivity even if locally repulsive.**

Conclusions and outlook

We wanted to understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

We conclude that at HF level, the non-local attraction:

- ◇ Acts as a source of Mott localisation;
- ◇ Enhances magnetisation at half filling;
- ◇ Can produce mixed-symmetry superconductivity;
- ◇ Can produce nematic superconductivity.

From here, we can follow various paths:

1. **Same model, other phases.**
Triplet superconductivity, complex breaking of translational invariance, nematic antiferromagnetism. . .
2. **Expand the model.**
Second nearest neighbours hopping, more refined potentials. . .
3. **More advanced numerical techniques.**
DMRG for chains and ladders; hybrid DMFT (local sector) and HF (non-local sector) to retrieve Mott physics induced by U . . .

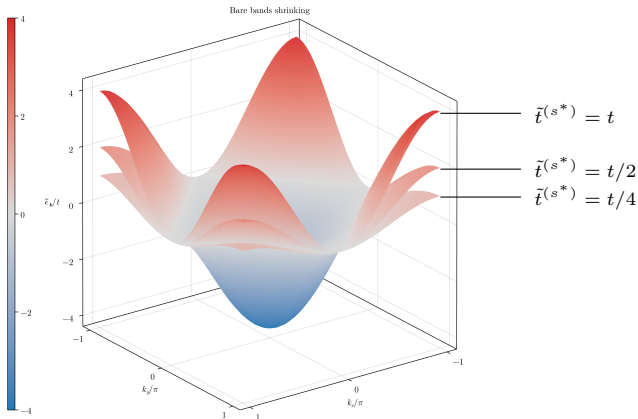
That's all, thank you!

Backup 1: details on Mott localisation

Bands flattening: a signature of Mott localisation (1/2)

Renormalisation of electronic bands, $\epsilon_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}\sigma}$:

$0 < \tilde{t}^{(s^*)} < t \implies$ **Flattening** of bands = Mott localisation.



Bands flattening: a signature of Mott localisation (2/2)

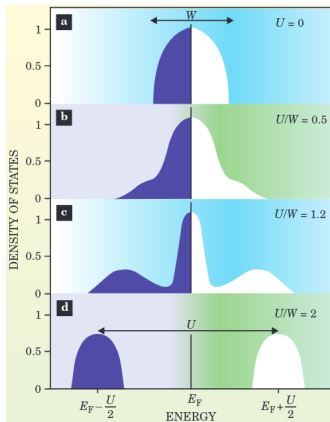


Figure 14: Image source: Gabriel Kotliar and Dieter Vollhardt, 2004. *Physics Today* 57 (March): 53–59.

As we increase the ratio U/t , the many-body DoS undergoes a transition:

- (a) Free model, conventional DoS;
- (b) Weak coupling, large coherent peak;
- (c) Intermediate coupling, **narrow coherent peak**;
- (d) Strong coupling, lower and upper Hubbard bands.



Localisation = flattening of bare bands.

Backup 2: details on the HF method

Hartree-Fock method: $\mathcal{S}$ is the set of all **gaussian states**, i.e. states that are non-interacting for some fermionic operators

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle, \quad \{\hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger}\} = \delta_{\alpha\beta},$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.



Wick's theorem: averages of quartic operators decompose nicely,

$$\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle = \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\delta} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\gamma} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\delta} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \langle \hat{c}_{\gamma} \hat{c}_{\delta} \rangle}_{\text{Cooper}}.$$

The EHM hamiltonian is a quartic operator!

The self-consistency mapping

Each $|\Psi\rangle \in \mathcal{S}$ is uniquely defined by a particular choice of the **variational parameters** (HFPs)

$$\Delta_{\alpha\beta}^{+-} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta} \rangle, \quad \Delta_{\alpha\beta}^{++} = \langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle.$$

Let $\mathbf{v}$ be the vector that collects all HFPs. We aim to find

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle,$$

with $\mathbf{v}$ such that:

Variational picture

Energy is minimised,

$$\left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi[\mathbf{v}]\rangle} = 0.$$

Self-consistency picture

A self-consistency problem is satisfied,

$$A(\mathbf{v}) = \mathbf{v},$$

being A a non-linear map **uniquely determined**.

The iterative algorithm (iHF)

The HF solution is the **fixed point** of a non-linear map A ,

$$A(\mathbf{v}) = \mathbf{v}.$$

$\Downarrow$

Strategy: “follow” the map up to the fixed point,

$$\mathbf{v}_{i+1} \equiv gA(\mathbf{v}_i) + (1 - g)\mathbf{v}_i.$$

with $i > 0$, $g \in [0, 1]$.

$\Downarrow$

Convergence criterion:

$$\forall j \quad : \quad [\mathbf{v}_{i+1} - \mathbf{v}_i]_j < \Delta v_j,$$

with $\Delta \mathbf{v}$ the vector of tolerances.

Example path (AF) ($t = 1.0$, $U = 3.0$, $V = 6.0$, $L = 128$, $\beta = 100.0$, $\delta = 0.0$)

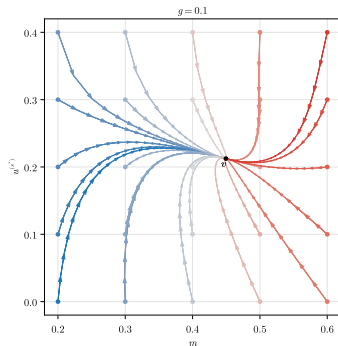


Figure 15: Example path for the AF phase.

Pros:

- ◇ Fully controllable;
- ◇ Easy trace-back of each effect;
- ◇ Computationally economic;
- ◇ Provides a framework to study phase competition;
- ◇ Modularity with other methods;
- ◇ A simple tool on top of a **much complex physics**.

Cons:

- ◇ Rough approximation of the real ground-state;
- ◇ Static;
- ◇ Phase competition must be understood with direct confrontation;
- ◇ Serial by construction.

Backup 3: Mott localisation at higher temperature

Temperature dependence of Mott effects in the normal phase

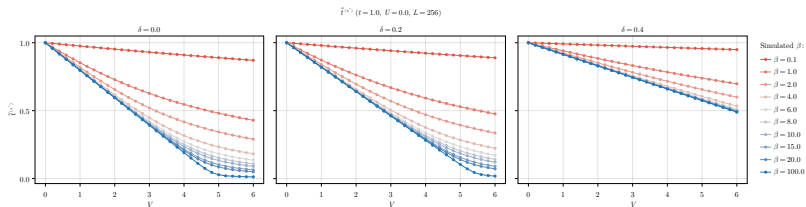


Figure 16: Comparative plots of $\tilde{t}^{(s^*)}$ in the normal phase at various temperatures and dopings.

In the **normal phase**:

Increase in temperature $\implies$ Disruption of Mott effects

Backup 4: Mott localisation in AF and SC phases

Effective hopping is:

- ◇ reduced by V ;
- ◇ **increased** by U .



The two interactions
“oppose” each other.

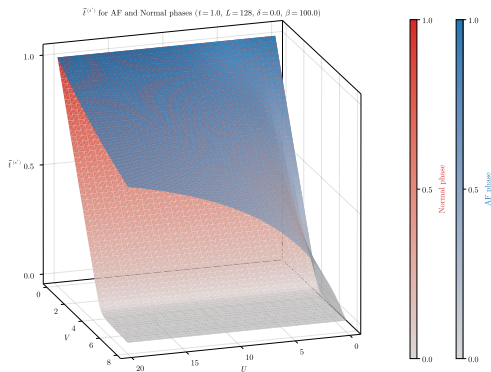


Figure 17: Comparative plot of $\tilde{t}^{(s^*)}$ for normal and AF phase.

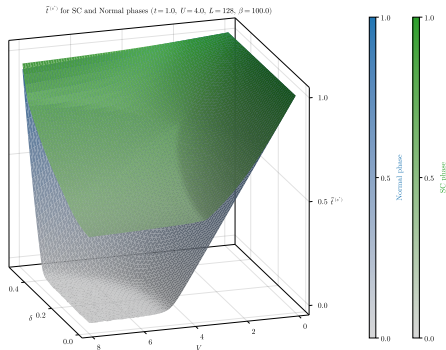


Figure 18: Comparative plot of $\tilde{t}^{(s^*)}$ for normal and AF phase.

Isotropic hopping
renormalisation:

$$\tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2}.$$

Formally identical to
normal and AF phases.

The effective hopping is mainly isotropic:

$$|\tilde{t}^{(s^*)}| \gg |\tilde{t}^{(d)}|.$$

Backup 5: energetic competition between AF and SC phases

The AF-SC phase boundary

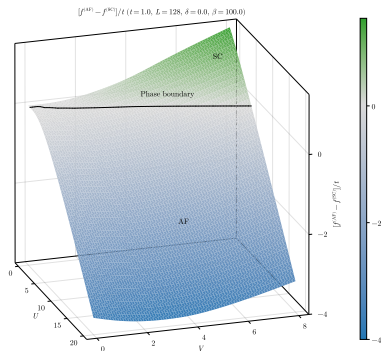


Figure 19: Energy difference: SC versus AF phase.

By computing the free energy difference,

$$f^{(\text{AF})} - f^{(\text{SC})},$$

we identify a phase boundary.
Possibly a critical ratio

$$\left(\frac{V}{U}\right)_c = k$$

exists?

The SC phase is dominant when $V \gg U, t$, the AF phase when $U \gg V, t$.

Backup 6: The origin of the $-V$ attraction

$$\hat{H}_V = -V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j : \quad \text{Isotropic, spin independent attraction.}$$

Postulated: originated by e -ph Coulomb interaction. It is believed that this attraction is mediated by phonons (lattice vibrations).

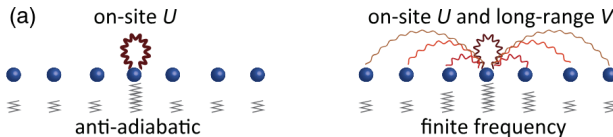


Figure 20: Image source: Wang et al., *Phys. Rev. Lett.* 127: 197003

Alternative origins? E.g. spin fluctuations?